

AI Job Agent: Master Architecture & Workflow Blueprint

End-to-End Autonomous Job Discovery, Multi-Tier Scoring, Dynamic ATS Tailoring & Dual-Stream Applications

Target Candidate: Kamal Kumar	Core Stack: Java, Spring Boot, Python, FastAPI, AWS, Docker, AI Agents
Target Experience: 0 to 2 Years ONLY (Strict Entry-Level)	Source Registry: 16 Active Connectors (APIs, ATS, Social Posts)
Scheduler Cycle: 24/7 Runs Every 6 Hours (06:00, 12:00, 18:00, 00:00 IST)	Frameworks: LangGraph + FastAPI + Gmail API + Gemini AI + openpyxl

1. Master Workflow & Pipeline Architecture

The system follows a clean, resilient sequential pipeline from multi-channel discovery to telemetry dispatch. Direct bot application on job boards is retired in favor of high-value Recruiter Email Outreach and 1-Click Application Link Preparation:



2. Dynamic Source Configuration (job_sources.yml)

Instead of hardcoding source logic in Python, the agent loads app/config/job_sources.yml at runtime. Sources can be toggled on/off and prioritized dynamically without code modifications:

```
# app/config/job_sources.yml
sources:
  greenhouse: { enabled: true, type: career_page, priority: high }
  lever: { enabled: true, type: career_page, priority: high }
  ashby: { enabled: true, type: career_page, priority: high }
  workday: { enabled: true, type: career_page, priority: high }
  smartrecruiters: { enabled: true, type: career_page, priority: high }
  jobicy: { enabled: true, type: remote, priority: high }
  remotive: { enabled: true, type: remote, priority: high }
  himalayas: { enabled: true, type: remote, priority: high }
  adzuna: { enabled: true, type: job_board, priority: high }
  agent_reach: { enabled: true, type: recruiter_post, priority: high }
  linkedin_posts: { enabled: true, type: recruiter_post, priority: high }
  ycombinator: { enabled: true, type: remote, priority: high }
  wellfound: { enabled: true, type: remote, priority: high }
  arbeitnow: { enabled: true, type: remote, priority: medium }
  weworkremotely: { enabled: true, type: remote, priority: medium }
  remoteok: { enabled: true, type: remote, priority: medium }
```

3. ExperienceAnalyzer & Weighted Multi-Dimensional Scoring

Dedicated ExperienceAnalyzer (Strict 0-2 Years Target):

The ExperienceAnalyzer inspects both the job title and full description. Any role requiring 3+ years, 4+ years, or ranges like 3–5 yrs / 2–4 yrs is rejected immediately before resume tailoring or application.

Requirement Pattern	Extracted Min / Max	Pipeline Action	Rationale & Behavior
Fresher / Entry / 0-2 yrs	Min: 0.0, Max: 2.0	PASS (Eligible)	Perfect target match. Advances to AI semantic scoring.
1-2 years / 2 years	Min: 1.0, Max: 2.0	PASS (Eligible)	Within candidate's 0-2 years background.
2-4 years / 2-5 years	Min: 2.0, Max: 4.0+	HARD REJECT	Exceeds 2-year upper threshold.
3+ years / 3-5 years	Min: 3.0, Max: 5.0	HARD REJECT	Experienced tier. Hard-blocked before LLM.
Senior / Staff / Lead / Mgr	Seniority Keyword	HARD REJECT	Senior title guard triggered. Score set to 0.

Multi-Dimensional Scoring Formula:

Overall Score = (30% × Role Match) + (25% × Experience Match) + (25% × Skill Match) + (10% × Location Match) + (10% × Freshness Quality)

4. Dual-Stream Application Strategy & Dynamic ATS Resumes

100% Unique Dynamic ATS Resumes:

For every qualifying job (Score ≥ 65), Gemini extracts key technologies and tailors the summary, skill badges, and work accomplishments. ReportLab Platypus builds a pixel-perfect, 1-page ATS compliant PDF saved to resumes/.

Application Stream	Target Sources	Execution Strategy & Safety Rules
📧 Stream A: Recruiter Outreach (REAL EMAIL)	LinkedIn Hiring Posts, HackerNews Threads, JD Recruiter Contacts	Zero Bounce Policy: Sends cold email outreach ONLY when explicit recruiter email is extracted from post/JD. Generates custom cover note and attaches tailored ATS PDF resume via Gmail API OAuth2.
🔗 Stream B: 1-Click Apply Link (APPLY LINK)	Job Boards (Adzuna, Naukri, Foundit, Shine, Hirst) & Company Career Portals	Direct Form Botting Removed: Eliminates fragile form botting and scraper bans. Generates tailored 1-page ATS PDF resume + customized cover letter and formats 1-click direct link in Excel report for fast user apply.

5. Production Codebase Directory Structure

Complete clean directory layout of the repository:

```
AI-Job-Agent/
├── app/
│   ├── agents/ # LangGraph Workflow Orchestrators
│   │   ├── nodes.py # 6 Pipeline Nodes (fetch, score, resume, apply, notify, report)
│   │   └── orchestrator.py # LangGraph StateGraph compilation
│   ├── state.py # AgentState TypedDict schema
│   ├── config/ # Central Configuration & Source Settings
│   │   ├── job_sources.yml # YAML source enable/disable & priority config
│   │   └── settings.py # Environment variables & candidate profile
│   ├── connectors/ # 16 Active Job Ingestion Connectors
│   │   ├── base.py # BaseConnector abstract interface
│   │   ├── manager.py # SearchManager concurrent executor
│   │   ├── registry.py # Connector discovery decorator
│   │   ├── greenhouse_connector.py # Greenhouse ATS API
│   │   ├── lever_connector.py # Lever ATS API
│   │   ├── ashby_connector.py # Ashby ATS API
│   │   ├── workday_connector.py # Workday CXS Portal
│   │   ├── smartrecruiters_connector.py # SmartRecruiters API
│   │   ├── jobcity_connector.py # Jobcity Remote API
│   │   ├── remotive_connector.py # Remotive Software API
│   │   ├── himalayas_connector.py # Himalayas Remote API
│   │   ├── adzuna_connector.py # Adzuna Jobs API
│   │   ├── agent_reach_connector.py # Agent Reach LinkedIn scraper
│   │   ├── linkedin_posts_connector.py # LinkedIn Hiring Posts parser
│   │   ├── ycombinator_connector.py # HackerNews / YC Algolia API
│   │   ├── wellfound_connector.py # Wellfound AI Startups
│   │   ├── arbeitnow_connector.py # Arbeitnow Tech API
│   │   └── weworkremotely_connector.py # WeworkRemotely RSS
│   ├── db/ # Database Storage Layer
│   ├── models.py # SQLAlchemy ORM models (Job, Application, Resume)
│   ├── session.py # SQLite engine (jobs.db)
│   ├── prompts/ # LLM System Personas & Prompts
│   │   ├── email_prompt.py # Cold outreach email prompt
│   │   ├── resume_prompt.py # Dynamic ATS resume tailoring prompt
│   │   └── scoring_prompt.py # Strict 0-2 yrs technical scoring prompt
│   ├── schemas/ # Data Transfer Objects
│   │   ├── application.py # Application status schema
│   │   └── job_data.py # Unified JobData dataclass
│   ├── services/ # Core Business Logic
│   │   ├── application_policy.py # Dual-Stream Policy Manager (Recruiter Email vs Apply Link)
│   │   ├── apply_service.py # Application routing & link preparation
│   │   ├── email_service.py # Gmail API OAuth2 sender
│   │   ├── mock_scoring_service.py # Heuristic pre-filter engine
│   │   ├── report_service.py # openpyxl multi-tab Excel generator
│   │   ├── resume_service.py # Dynamic ATS 1-page PDF builder
│   │   └── scheduler_service.py # APScheduler 24/7 6-hour cron service
│   ├── scoring_service.py # Gemini AI semantic scoring
│   ├── whatsapp_service.py # CallMeBot & Meta WhatsApp alerts
│   ├── utils/ # Validators & Helper Utilities
│   │   ├── email_validator.py # Explicit HR email parser & DNS validator
│   │   ├── experience_analyzer.py # Dedicated 0-2 yrs ExperienceAnalyzer
│   │   ├── experience_filter.py # Senior title & experience hard guard
│   │   └── gemini_client.py # Resilient Gemini client with retry
│   ├── data/ # Persistent Storage
│   │   ├── jobs.db # SQLite primary database
│   │   └── docs/ # Architecture Documentation & PDF Blueprints
│   ├── AI-Job-Agent-Architecture-and-Workflow.pdf
│   ├── reports/ # Daily Multi-Tab Excel Reports
│   ├── resumes/ # Master & Tailored ATS PDF Resumes
│   ├── scripts/ # Operational Scripts & Tools
│   ├── run_agent.py # CLI runner for single pipeline execution
│   ├── setup_gmail_token.py # Gmail OAuth2 token generator
│   ├── setup_whatsapp.py # WhatsApp CallMeBot diagnostic tool
│   ├── generate_architecture_pdf.py # Architecture PDF builder
│   ├── main.py # FastAPI 24/7 Server & Background Scheduler Entrypoint
│   └── requirements.txt # Production Python dependencies
├── .env # Configuration & API Credentials
```